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Causal Models in Data Science

Final Project Report

Causal Question and Motivation

Causal Question: What is the causal effect of online advertising exposure on user conversion?

Motivation:
Measuring advertising effectiveness is one of the most challenging problems in digital marketing. A naive comparison of conversion rates between users who saw ads versus those who didn't produces biased estimates because these groups differ systematically. Users who see ads tend to be more engaged, have higher purchase intent, or use devices more conducive to ad delivery. This selection bias leads advertisers to overestimate the true impact of their campaigns.

Addressing this problem requires causal inference methods that can isolate the effect of ad exposure from confounding factors. Understanding the true causal effect, and how it varies across users, is critical for optimizing advertising spend, improving targeting strategies, and accurately measuring return on investment.

Context:
This analysis uses the Criteo Uplift Modeling Dataset (Diemert et al., 2018), which contains approximately 14 million observations from real-time bidding (RTB) advertising campaigns. The dataset features:

- **Treatment:** Randomized assignment to participate in RTB auctions
- **Exposure:** Whether the user actually saw an advertisement (subject to non-compliance)
- **Outcome:** Binary conversion indicator (purchase or signup)
- **Covariates:** 12 anonymized user features (f0-f11)

The randomized treatment assignment combined with realistic non-compliance makes this dataset ideal for applying instrumental variables and other causal inference methods.

Roadmap and Key Findings
<u>Roadmap</u>

1. **Exploratory Data Analysis:** Data source and variable descriptions
2. **Causal Identification and Estimation:** Identification assumptions and implementation of two causal inference methods (IV/2SLS and DML-IV)
3. **Results and Comparative Analysis:** Point estimates, confidence intervals, and method comparison
4. **Evaluation:** Sensitivity analysis and robustness assessment using refutation tests
5. **Conclusion and Future Work:** Summary of findings, limitations, and next steps

Key Findings

- **IV/2SLS:** Ad exposure increases conversion probability by 2.82 percentage points for compliers (95% CI: 2.63–3.01 pp, N = 14M)
- **DML-IV:** Ad exposure increases conversion probability by 2.99 percentage points for compliers (95% CI: 2.98–3.00 pp, N = 500K)
- **Consistency:** Both methods yield consistent estimates (~2.8–3.0 pp), suggesting linear covariate adjustment in 2SLS is adequate
- **Robustness:** All three refutation tests pass (placebo treatment, data subset, random common cause)
- **Interpretation:** Ads roughly double conversion probability for compliers (from ~2.6% to ~5.4%)

Exploratory Data Analysis

Data Source: [description link, data link]
This analysis uses the Criteo Uplift Modeling Dataset (Diemert et al., 2018), a publicly available dataset containing approximately 14 million observations from real-time bidding (RTB) advertising campaigns. The dataset was constructed from multiple incrementality tests with randomized treatment assignment.

Variable Description:

Variable	Type	Role	Description
f0-f11	Continuous	Covariates	12 anonymized user features
treatment	Binary	Instrument (Z)	Randomized assignment to RTB auction
exposure	Binary	Treatment (D)	Whether user saw an ad
conversion	Binary	Outcome (Y)	Whether user converted
visit	Binary	Outcome (secondary)	Whether user visited

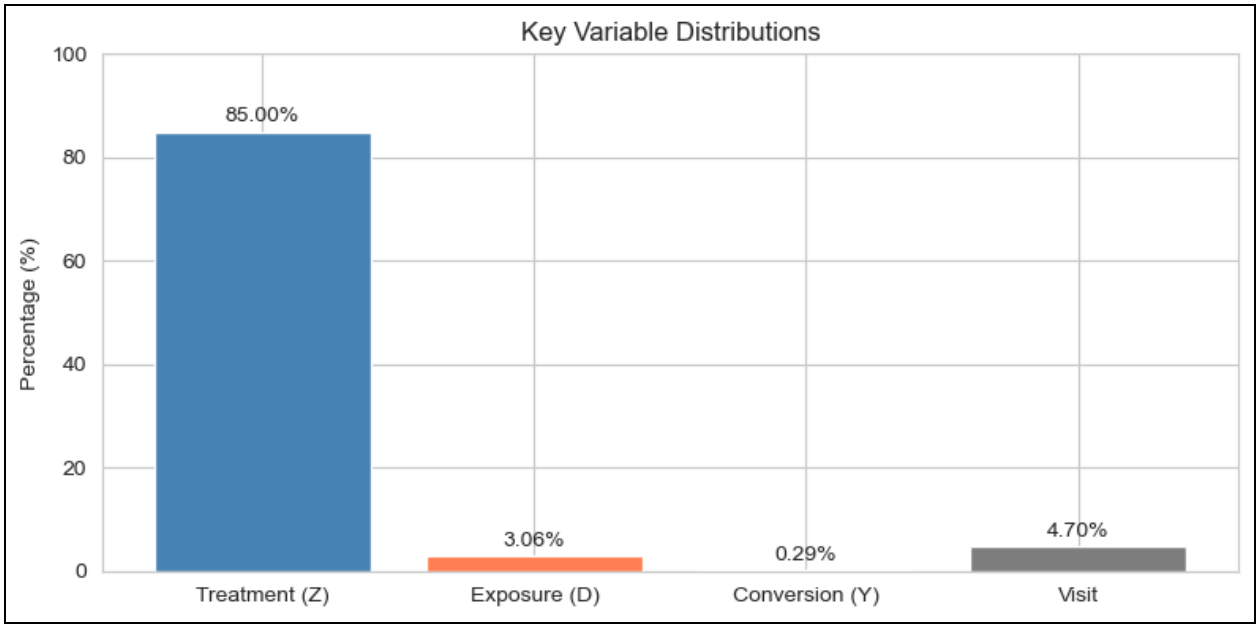

Figure 1. Distribution of key binary variables. 85% of users were assigned to treatment, 3.1% were exposed to ads, 0.29% converted, and 4.7% visited.

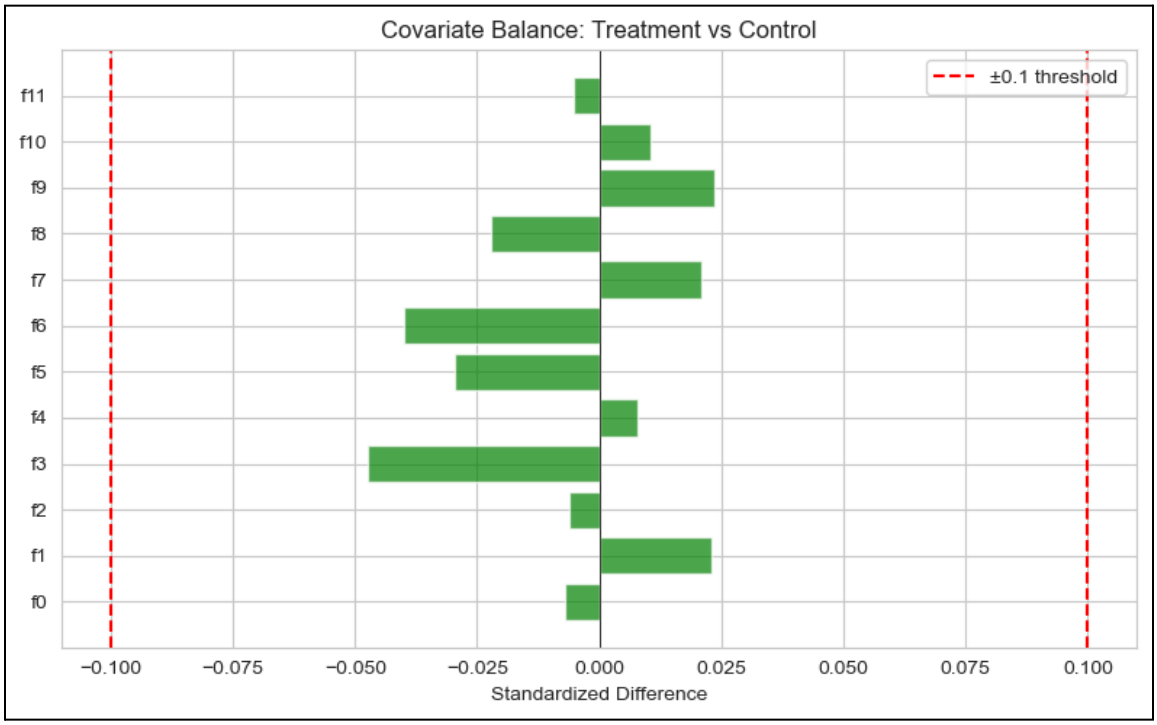

Figure 2. Covariate balance between treatment and control groups. All standardized differences fall within the ±0.1 threshold, confirming successful randomization.

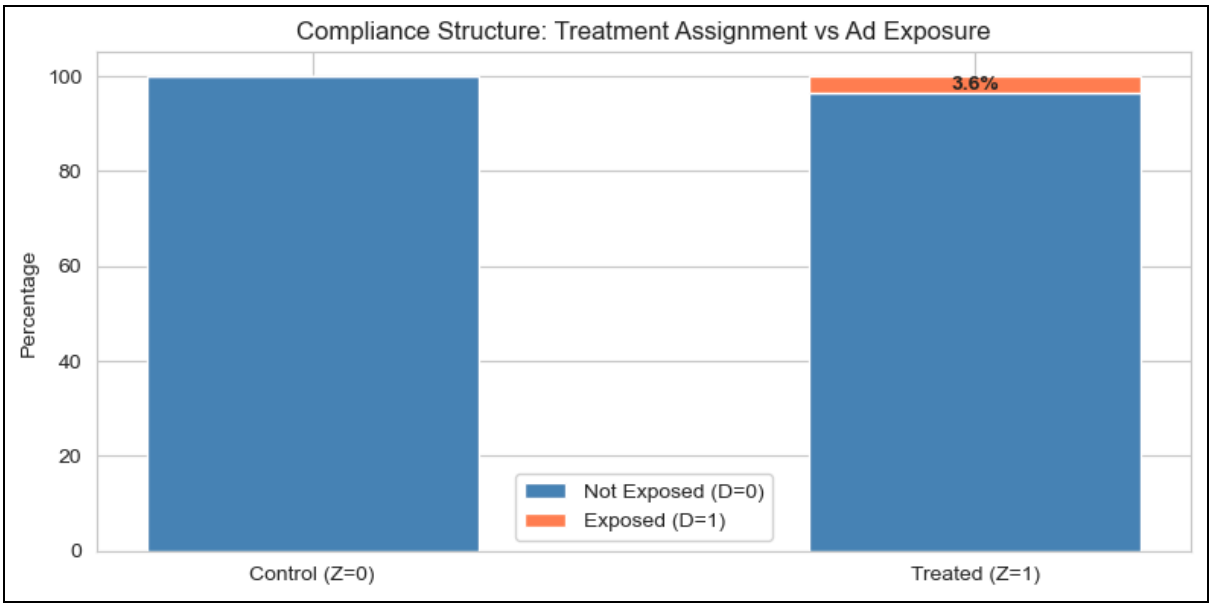

Figure 3. Compliance structure showing treatment assignment versus ad exposure. Control users (Z=0) were unexposed, while only 3.6% of treated users (Z=1) were exposed. This reveals one-sided non-compliance: 3.6% compliers, 96.4% never-takers, and no always-takers or defiers.

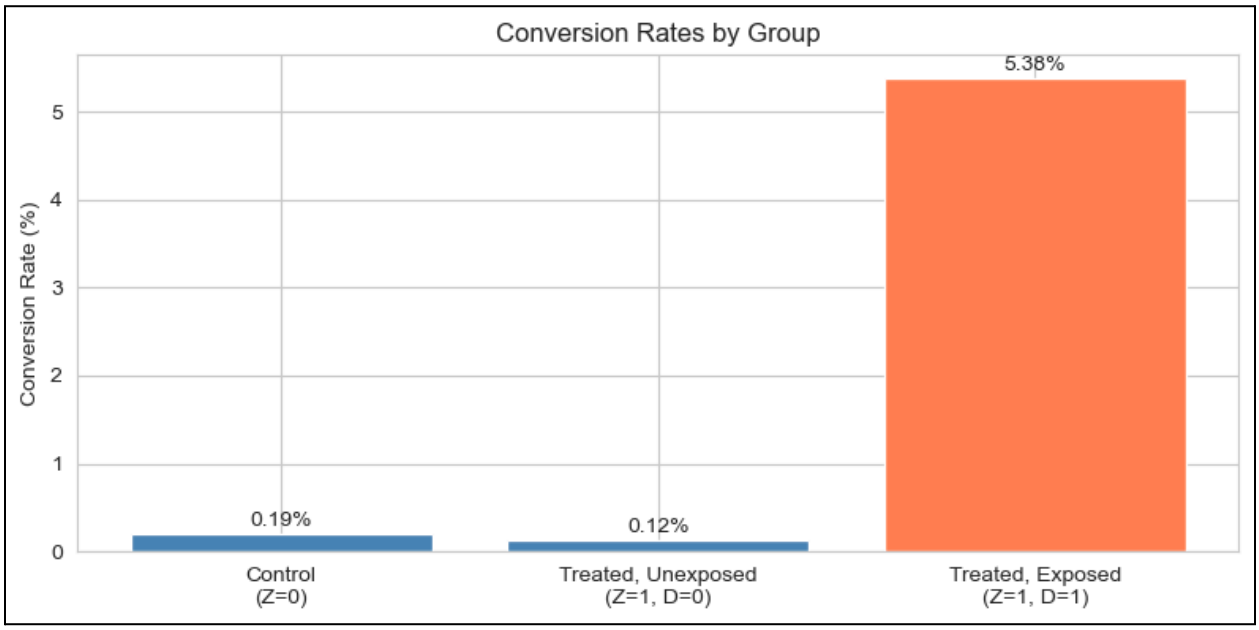

Figure 4. Conversion rates by treatment and exposure status. Exposed users convert at 5.38% vs 0.19% (control) and 0.12% (treated, unexposed). This gap reflects both causal effect and selection bias.

Causal Identification and Estimation

Identification Assumptions:

The causal structure is represented by the following diagram:

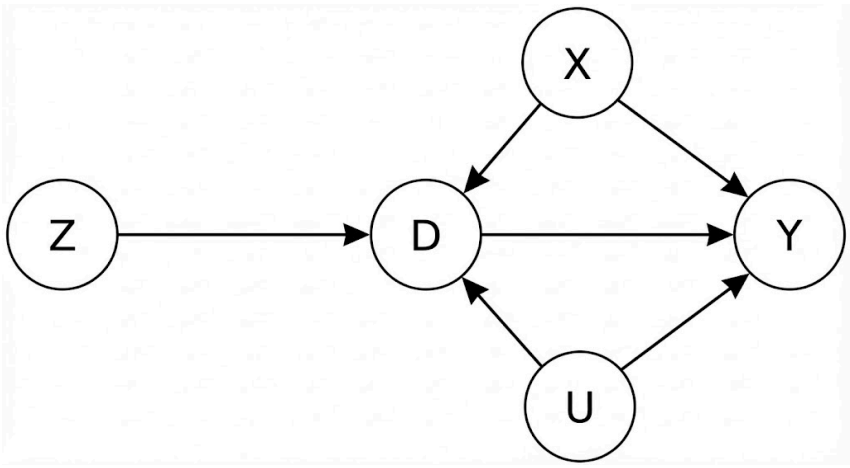

Figure 5. Causal graph. Randomized assignment (Z) affects conversion (Y) only through exposure (D). X represents observed covariates; U represents unobserved confounders addressed by IV.

Assumption	Description	Justification in Data
SUTVA	No interference between users; no hidden treatment versions	Users are independent; one user's ad doesn't affect another's conversion
Relevance	Instrument (Z) affects treatment (D)	Verified: Z strongly predicts D (3.6% compliance rate, huge sample = strong first stage)
Exogeneity	Instrument is as-if random	Z is randomized by design
Exclusion Restriction	Z affects Y only through D	Assignment to auction doesn't directly cause conversion; only seeing the ad does
Monotonicity	No defiers	Verified: 0% of control users were exposed, confirming no defiers exist.

Causal Methods:

Method 1: Instrumental Variables (IV/2SLS)

Method Overview

Ad exposure is not randomly assigned, so comparing exposed and unexposed users conflates causal effects with selection bias. We address this using Instrumental Variables (IV) estimated via Two-Stage Least Squares (2SLS), with randomized treatment assignment (Z) as the instrument for exposure (D). This approach recovers the Local Average Treatment Effect (LATE), the causal effect for compliers whose exposure depends on their assignment.

$$\text{LATE} = \mathbb{E}[Y_1 - Y_0 \mid \text{Compliers}]$$

Implementation

The first stage regresses exposure on assignment and covariates (D ~ Z + X), yielding an F-statistic of 76,419, far exceeding the threshold of 10 and confirming instrument strength:

$$\text{First Stage:} \quad D = \alpha_0 + \alpha_1 Z + \alpha X + \varepsilon$$

The second stage regresses conversion on predicted exposure and covariates, where the coefficient on $\hat{D}$ is the LATE estimate (Y ~ $\hat{D}$ + X):

$$\text{Second Stage:} \quad Y = \beta_0 + \beta_1 \hat{D} + \beta X + \nu$$

Results & Interpretation

The coefficient on $\hat{D}$ is 0.0282 (β_1 = 0.0282), meaning ad exposure **increases conversion probability by 2.82 percentage points** for compliers (95% CI: 2.63–3.01 pp, $p < 0.001$). Compliers, the 3.6% of users whose exposure depends on assignment, convert at 5.38% when exposed. Without exposure, their implied conversion rate is 2.56%, meaning ads roughly double their conversion probability.

Method 2: Double Machine Learning with IV (DML-IV)

Method Overview

2SLS assumes linear relationships between covariates and outcomes. Double Machine Learning (DML) relaxes this by using flexible machine learning models to control for covariates while preserving valid causal inference. We use DML with instrumental variables, maintaining the IV framework while allowing nonlinear covariate adjustment.

Implementation

We fit DML-IV on a random sample of 500K observations. The first step residualizes each variable by predicting it from covariates using Random Forest and subtracting the prediction:

$$\tilde{Y} = Y - \mathbb{E}[Y \mid X]$$
$$\tilde{D} = D - \mathbb{E}[D \mid X]$$
$$\tilde{Z} = Z - \mathbb{E}[Z \mid X]$$

The second step applies cross-fitting (5-fold), where predictions are made on held-out data to prevent overfitting bias.

The third step performs IV regression on residualized variables, where the coefficient on $\tilde{D}$ is the LATE:

$$\tilde{Y} = \beta_1 \tilde{D} + \nu$$

Results & Interpretation

The coefficient β_1 = 0.0299, meaning ad exposure **increases conversion probability by 2.99 percentage points** for compliers (95% CI: 2.98–3.00 pp). Compliers, the 3.6% of users whose exposure depends on assignment, convert at 5.38% when exposed. Without exposure, their implied conversion rate is 2.39%, meaning ads more than double their conversion probability.

Results and Comparative Analysis

Estimated Effects:

Method	LATE (pp)	95% CI
IV/2SLS	2.82	[2.63, 3.01]
DMV-IV	2.99	[2.98, 3.00]

Method Comparison:

Both methods estimate the Local Average Treatment Effect (LATE) for compliers, the 3.6% of users whose ad exposure depends on treatment assignment. IV/2SLS yields 2.82 pp while DML-IV yields 2.99 pp, a difference of 0.17 pp. The DML-IV estimate falls within the IV/2SLS confidence interval, indicating statistical consistency.

The similarity suggests that linear covariate adjustment in 2SLS is adequate for this data. The slightly higher DML estimate may reflect flexible adjustment for nonlinearities in covariates, though the difference is small relative to the effect size. The consistency across methods strengthens our confidence in the causal interpretation: ad exposure approximately doubles conversion probability for compliers (from ~2.6% to ~5.4%).

Evaluation

We validate our causal estimates using DoWhy's refutation tests on a random sample of 500K observations. These tests stress-test whether the effect is genuine or an artifact of the method by probing different potential failures of our identification strategy.

Test	Description	New Effect	p-value	Result
Placebo Treatment	Replace treatment with random noise	~0	0.0	Pass
Data Subset	Re-estimate on 80% of data	0.0345	0.92	Pass
Random Common Cause	Add simulated unobserved confounder	0.0347	1.0	Pass

All three tests pass, strengthening confidence in the causal interpretation.

The placebo treatment test replaces actual exposure with random noise. The effect disappears (new effect ≈ 0), confirming that our estimate reflects a genuine treatment-outcome relationship rather than spurious correlation. The low p-value indicates the original estimate is significantly different from the placebo.

The data subset test re-estimates the effect on a random 80% of observations. The estimate remains stable (0.0345 vs 0.0347), with a high p-value (0.92) indicating no significant difference. This confirms robustness to sampling variation.

The random common cause test adds a simulated unobserved confounder to the model. The estimate is unchanged ($p = 1.0$), suggesting robustness to potential unobserved confounding. Combined with our instrumental variables approach, this reinforces the validity of our causal identification.

Conclusion and Future Work

Summary of Findings:

This study estimates the causal effect of online ad exposure on user conversion using the Criteo Uplift Modeling Dataset (14 million users). Using randomized treatment assignment as an instrument for ad exposure, we estimate a Local Average Treatment Effect (LATE) of approximately 2.8–3.0 percentage points for compliers. IV/2SLS yields 2.82 pp (95% CI: 2.63–3.01), while DML-IV yields 2.99 pp (95% CI: 2.98–3.00). For compliers, the 3.6% of users whose ad exposure depends on assignment, ads roughly double conversion probability (from ~2.6% to ~5.4%). Refutation tests confirm the robustness of these estimates.

Limitations:

Several limitations warrant consideration. First, the LATE applies only to compliers as effects for always-takers and never-takers remain unidentified. Second, while we control for 12 observed covariates, unobserved confounders may still bias estimates if the exclusion restriction is violated. Third, computational constraints limited DML-IV and refutation tests to 500K observations; estimates may differ slightly with the full dataset. Fourth, the binary conversion

outcome does not capture visit frequency or engagement depth, which may be more policy-relevant for advertisers. Finally, the anonymized covariates limit interpretability of heterogeneous effects.

Future Work:

1. **Alternative outcomes:** Use visit data as a continuous outcome to capture engagement beyond binary conversion.
2. **Heterogeneous effects:** Apply Causal Forest or Instrumental Forest to estimate conditional average treatment effects (CATE) across user segments.
3. **Ground truth evaluation:** If individual treatment effects were available, evaluate CATE estimates using PEHE (Precision in Estimation of Heterogeneous Effects), the RMSE between estimated and true individual effects.
4. **Alternative ML models:** Explore LightGBM, XGBoost, or neural networks as nuisance function learners in DML to assess sensitivity to model choice.
5. **Full-scale estimation:** Run DMV-IV and refutation tests on the full 14 million observations with additional computational resources to obtain more precise estimates.

Link to the code repository:

https://github.com/mokhan95/causal-models-final-project